Psychological Testing (with Practicum)

Course Syllabus

Instructor: Hau-Hung Yang

Course Information

Course Level:	Undergraduate (upper level)
Credits:	3-credit lecture + 1-credit companion practicum
Duration:	One semester
Proposed Hours:	3 lecture hours + 2 practicum hours per week
Prerequisites:	Introductory statistics for psychology
Software:	R (packages: `psych`, `mirt`, `lavaan`, `CTT`)

Course Description

This course introduces the theory and practice of psychological testing: reliability and validity, test construction and evaluation, and score interpretation under classical test theory (CTT) and item response theory (IRT). The 3-credit lecture is paired with a separately assessed, 1-credit **practicum**. Weekly labs cover administration and scoring with permitted materials, item and reliability analyses in R, fairness across groups, and test reports. Public or simulated data are available for every project. Responsible test use in research, education, clinical, and health settings is emphasized throughout.

Course Objectives

Upon completion of this course, students will be able to:

1. Explain the core concepts of measurement: constructs, operationalization, scales, and norms.
2. Estimate and interpret reliability (internal consistency, test–retest, inter-rater) and validity (content, criterion, construct) evidence.
3. Conduct item analysis (difficulty, discrimination, distractor analysis) under both CTT and IRT.
4. Administer, score, and interpret standardized tests following ethical and professional standards.

5. Evaluate measurement fairness, including differential item functioning (DIF) across groups.
6. Produce a clear, well-justified psychological test report from raw data.

Textbooks and References

- Furr, R. M. (2021). *Psychometrics: An Introduction* (4th ed.). SAGE. **[Primary textbook]**
- Cohen, R. J., Schneider, W. J., & Tobin, R. M. (2022). *Psychological Testing and Assessment* (10th ed.). McGraw-Hill.
- AERA, APA, & NCME (2014). *Standards for Educational and Psychological Testing*.
- de Ayala, R. J. (2022). *The Theory and Practice of Item Response Theory* (2nd ed.). Guilford. [Supplementary]

Assessment

The lecture and companion practicum receive separate grades.

Lecture (3 credits)	
Midterm examination	35%
Final examination	35%
Written critiques of measurement evidence	20%
Participation	10%
Practicum (1 credit)	
Lab assignments (administration, scoring, R analyses)	40%
Test-report project, including AI-analysis critique and presentation	50%
Lab attendance and participation	10%

Weekly Schedule (Lecture + Practicum)

Week	Topic (Lecture + *Practicum*)
1	Introduction to measurement: constructs, test types, and applications; informed consent and confidentiality before any practice administration. *Practicum: lab orientation and R setup*
2	Norms, standardization, and standard scores. *Practicum: permitted test materials, de-identification, and scoring a sample instrument*

Week	Topic (Lecture + *Practicum*) (continued)
3	Classical test theory and reliability I: true score, error, and internal consistency (Cronbach's α, McDonald's ω). *Practicum: reliability analysis in R with the* `psych` *package*
4	Reliability II: test–retest, parallel forms, inter-rater reliability; the standard error of measurement. *Practicum: test–retest and inter-rater reliability in R*
5	Validity I: content and criterion-related validity. *Practicum: criterion validity with a real dataset*
6	Validity II: construct validity, convergent/discriminant evidence; factor analysis. *Practicum: exploratory factor analysis in R.* First lecture critique due
7	Item analysis under CTT: item difficulty, discrimination, and distractor analysis. *Practicum: item analysis lab*
8	*Midterm Examination*
9	Introduction to item response theory: the 1PL/2PL models and item characteristic curves. *Practicum: plotting item characteristic curves in R*
10	IRT item and test information; ability estimation. *Practicum: fitting an IRT model with* `mirt`
11	Test construction and item banking: blueprint, item writing, pilot testing, and selection. *Practicum: short-scale analysis with public/simulated data or an instructor-approved pilot*
12	Test fairness and bias: differential item functioning (DIF). *Practicum: DIF detection in R.* Second lecture critique due
13	Applications in clinical, health, and educational assessment; computerized and adaptive testing overview. *Practicum: project analysis and consultation*
14	Ethical score interpretation and report writing. *Practicum: draft report; compare an AI-generated analysis with one's own, check its reasoning, and correct unsupported claims*
15	Project presentations: test-construction / test-report projects (lecture and practicum sessions)
16	*Final Examination*

The schedule follows the 16-week NTU semester: fourteen teaching weeks, the midterm examination in Week 8, and the final examination in Week 16. A class lost to a national holiday is made up in the flexible week after Week 16 (Week 16+1), as the NTU academic calendar allows.

Written Assignments

Lecture critiques. Submit two 1–2 page evidence reviews using instructor-provided studies or test materials, each worth 10% of the lecture grade. The first, due in Week 6, evaluates a validity claim; the second, due in Week 12, evaluates a fairness claim. Grading considers accurate identification of the claim and evidence, recognition of limitations, and clear justification of the conclusion.

Practicum project. Each student (or pair) will construct a short scale or evaluate a publicly available instrument. Public or simulated data may be used; any participant recruitment or pilot administration requires the instructor's prior approval of consent, confidentiality, and data handling. The report (approximately 10–15 pages) should include the construct definition, item selection, reliability and validity evidence, item analysis, a fairness check where feasible, and limitations. Submit reproducible R code and a brief critique of an AI-generated analysis, identifying and correcting any unsupported interpretation; the instructor can supply the AI output. This critique is part of the project grade. Drafts receive feedback in Week 14; final reports and presentations are due in Week 15.

Course Policies

- Practicum attendance is required; test materials and data must be handled ethically and confidentially.
- All lab work and the project must be submitted as reproducible R scripts or R Markdown documents.
- Late submissions incur a 10% penalty per day, up to three days.
- AI assistance must be disclosed and checked independently. Do not upload confidential data or restricted test materials to external AI tools; instructor-provided outputs can be used for the critique.